		Maximum observable is 3rd order. Therefore, answer is 7.
23	A	For one of the bright light, the first order minima is located at an angle θ where $$sin\theta = \frac{\lambda}{0.004}$$ If the central maxima of the second bright line is to coincide with the first order minima of the first bright light (Rayleigh Criterion for resolution of image), the second bright line must be at the same angle hence, $$tan\theta = \frac{d}{4 \times 10^5}$$ If the angle is small, we may approximate $tan\theta = sin\theta$ $$\frac{d}{4 \times 10^5} = \frac{\lambda}{0.004}$$ For the minimum *d*, we choose the smallest visible wavelength of light ie approximately 400 nm, Hence $d = 40$
24	B	e.m.f. of cell, *E* = 1.5 V (From Fig. a)